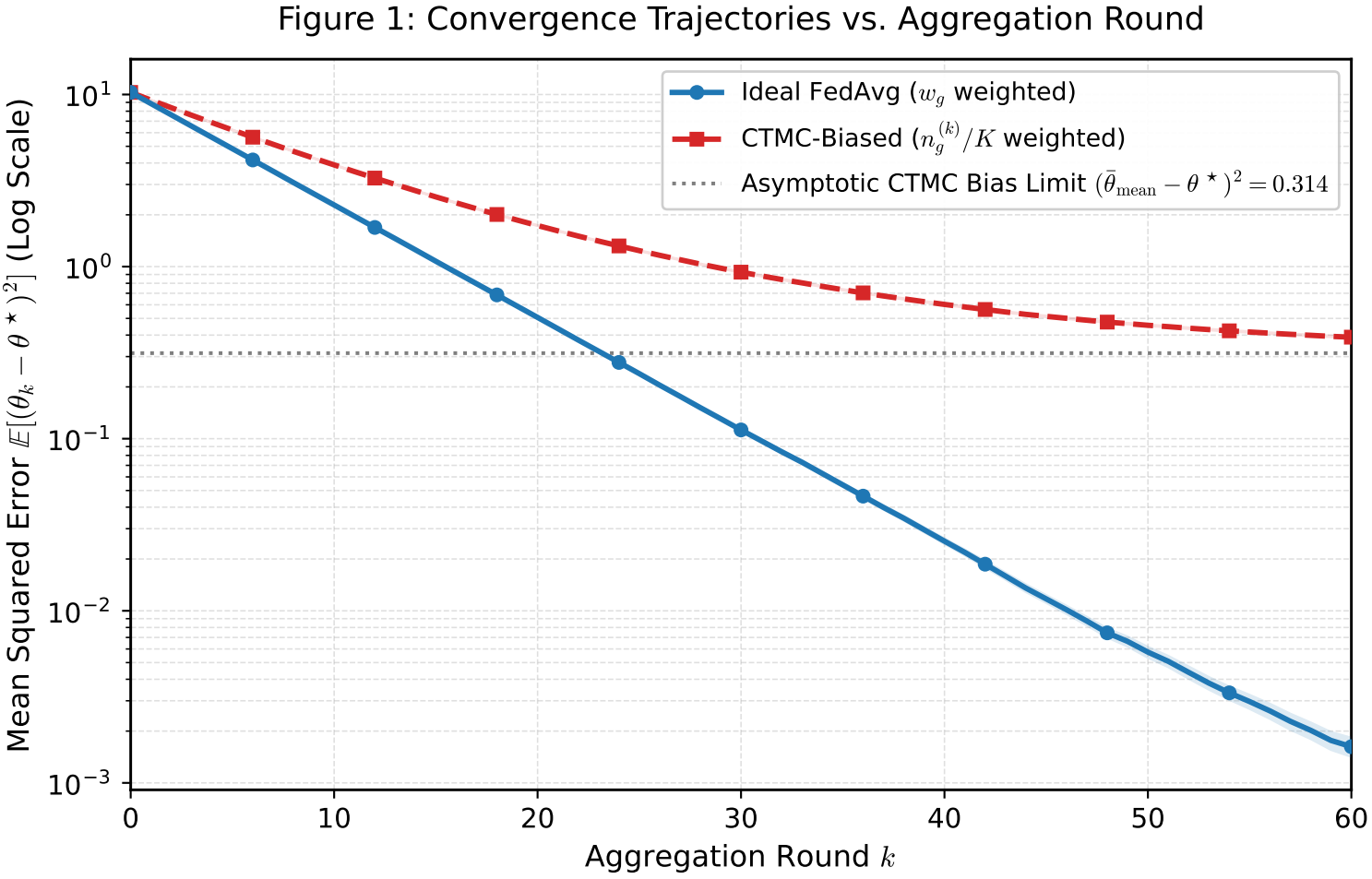
Figure 1: Convergence Trajectories vs. Aggregation Round
Ideal FedAvg (w_g weighted)
CTMC-Biased ($n_g^{(k)}/K$ weighted)
Asymptotic CTMC Bias Limit $(\bar{\theta}_{\text{mean}} - \theta^\star)^2 = 0.314$
Mean Squared Error $\mathbb{E}[(\theta_k - \theta^\star)^2]$ (Log Scale)
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
10
20
30
40
50
60
Aggregation Round k